Neural-Driven Computing

Advanced Systems Engineering (ASE2026)

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Research topics

- Have you ever wondered why we need a full stack of Operating System and a Browser to run AI workloads?
- Can't we just use LLMs to create "runnable" perceptrons instead of writing tons of code with it?
- Is it possible to use Perceptron-based computing as an operating system?

Why this matters

- Current AI mostly requires heavy OS + Browser stack
- Neural networks are universal function approximators - why not use them directly?
- Perceptron-based computing could eliminate the "abstraction tax" of traditional OS
- Enables AI-native hardware without legacy baggage
- Could unlock orders-of-magnitude efficiency gains for specialized workloads

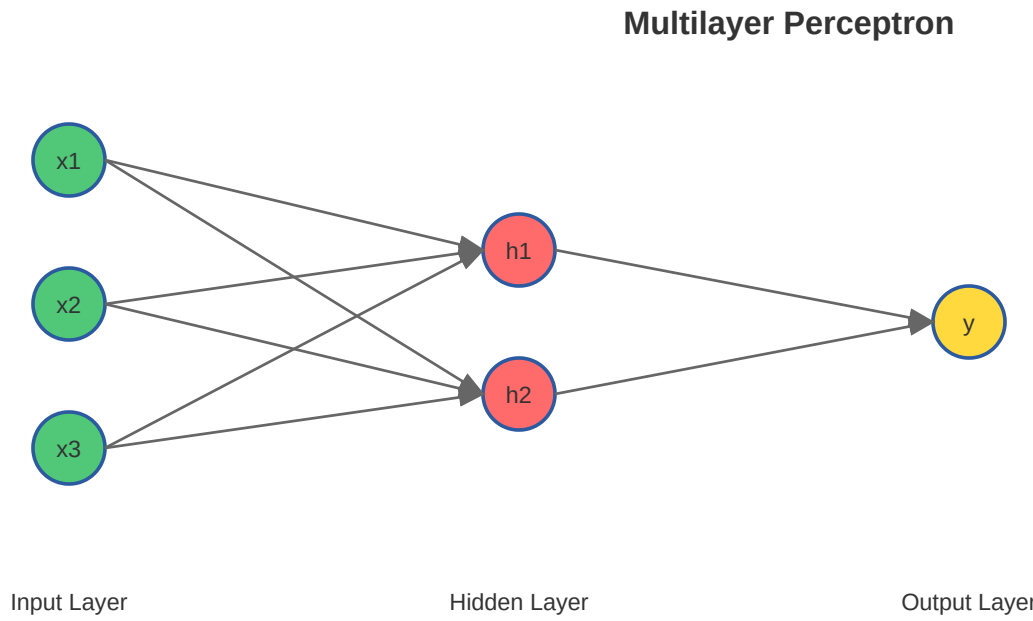
Where we come from

- 1940s-1950s: Punch Cards & Batch Processing
- 1960s: Mainframe OS & Time-Sharing
- 1970s-1980s: Personal Computers & GUIs - CP/M (1974) for microcomputers.
- 1990s: Windows & Linux - (GUI and multitasking)
- 2000s-2010s: Mobile OS with Browsers
- 2020s-now: AI all over but not for Operating Systems

What an operating system does

- Abstracts the hardware
- Memory management
- I/O handling
- Allows multiple processes
- Multiple users
- Tries hard to separate processes and users from each other

What a multilayer network does



z1 = w11*x1 + w12*x2 + w13*x3 + b1

h1 = ReLU(z1) ReLU

z2 = w21*x1 + w22*x2 + w23*x3 + b2

h2 = ReLU(z2)

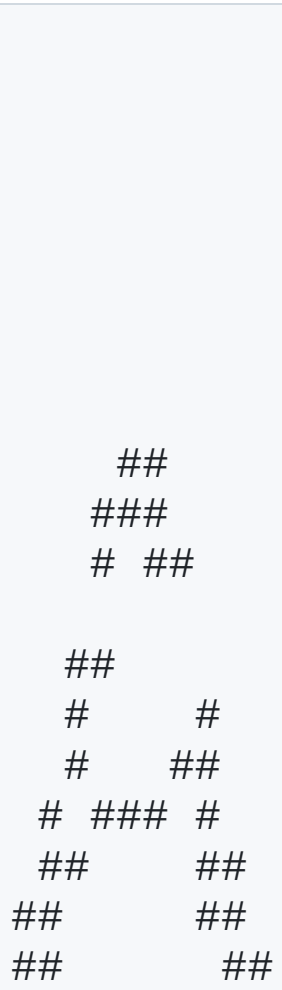
z3 = w31*h1 + w32*h2 + b3

y = softmax(z3) softmax

Approach

- RISC-V + Tensor OPs [✓]
- Emulator + RTL (system verilog) [✓]
- Assembler for the extensions [✓]
- Minimalistic bootloader [✓]
- Pytorch for training [✓]
- A "transpiler" for loadable and runnable artifact out of the trained network [✓]

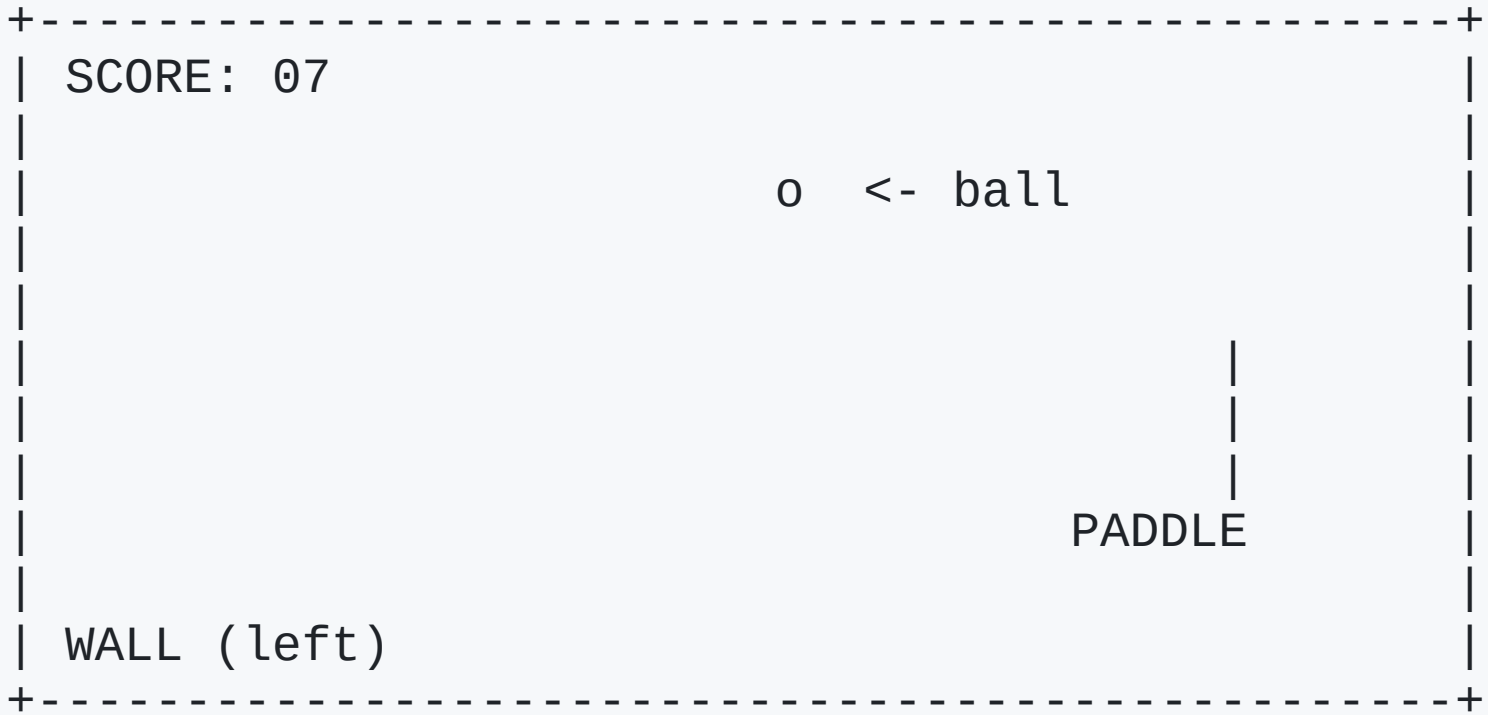
Simple character input [0..255] to framebuffer [20x20] output



Moveable character from [h,j,k,l] input in a [20x20] framebuffer

```
#
```

Future work - squash mockup



Future work - pong over network

- Dual-emulator networking
- connect two emulator instances and exchange state/tokens each frame.
- multiplayer pong (left/right paddles, synchronized ball state).

Thank you

Thanks for your attention I hope you've enjoyed the presentation!
Feel free to ask me anything.